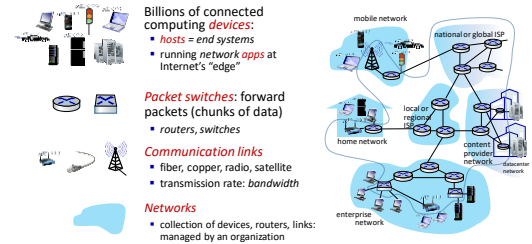


Internet Architecture



The Internet: a “nuts and bolts” view



“Fun” Internet-connected devices



The Internet: a “nuts and bolts” view

• Internet: “network of networks”

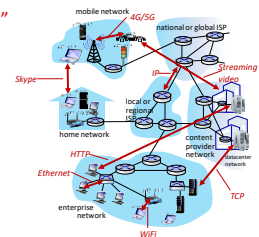
- Interconnected ISPs

• protocols are everywhere

- control sending, receiving of messages
- e.g., HTTP (Web), streaming video, Skype, TCP, IP, WiFi, 4G, 5G, Ethernet

• Internet standards

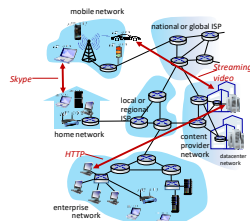
- RFC: Request for Comments
- IETF: Internet Engineering Task Force



The Internet: a “services” view

• **Infrastructure** that provides services to applications:

- Web, streaming video, multimedia teleconferencing, email, games, e-commerce, social media, inter-connected appliances, ...



What’s a protocol?

Human protocols:

- “what’s the time?”
- “I have a question”
- introductions

Network protocols:

- computers (devices) rather than humans
- all communication activity in Internet governed by protocols

Rules for:

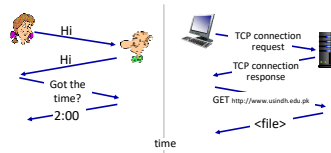
... specific messages sent

... specific actions taken when message received, or other events

Protocols define the format, order of messages sent and received among network entities, and actions taken on message transmission, receipt

What's a protocol?

A human protocol and a computer network protocol:

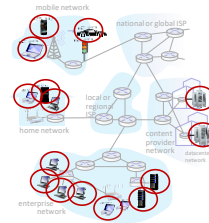


Q: other human protocols?

A closer look at Internet structure

Network edge:

- hosts: clients and servers
- servers often in data centers



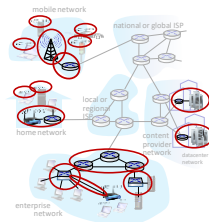
A closer look at Internet structure

Network edge:

- hosts: clients and servers
- servers often in data centers

Access networks, physical media:

- wired, wireless communication links



A closer look at Internet structure

Network edge:

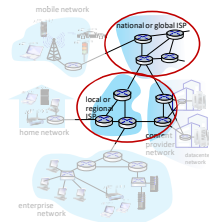
- hosts: clients and servers
- servers often in data centers

Access networks, physical media:

- wired, wireless communication links

Network core:

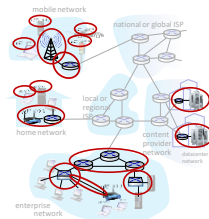
- interconnected routers
- network of networks



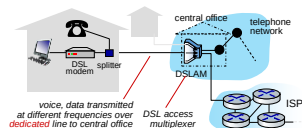
Access networks and physical media

Q: How to connect end systems to edge router?

- residential access nets
- institutional access networks (school, company)
- mobile access networks (WiFi, 4G/5G)

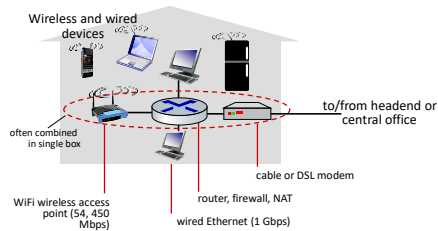


Access networks: digital subscriber line (DSL)



- use **existing** telephone line to central office DSLAM
 - data over DSL phone line goes to Internet
 - voice over DSL phone line goes to telephone net
- 24-52 Mbps dedicated downstream transmission rate
- 3.5-16 Mbps dedicated upstream transmission rate

Access networks: home networks



Wireless access networks

Shared *wireless* access network connects end system to router

- via base station aka "access point"

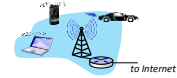
Wireless local area networks (WLANs)

- typically within or around building (~100 ft)
- 802.11b/g/n (WiFi): 11, 54, 450 Mbps transmission rate

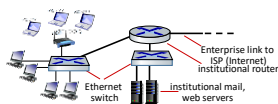


Wide-area cellular access networks

- provided by mobile, cellular network operator (10's km)
- 10's Mbps
- 4G cellular networks (5G coming)



Access networks: enterprise networks



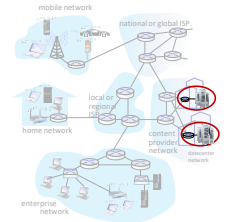
- companies, universities, etc.
- mix of wired, wireless link technologies, connecting a mix of switches and routers (we'll cover differences shortly)
 - Ethernet: wired access at 100Mbps, 1Gbps, 10Gbps
 - WiFi: wireless access points at 11, 54, 450 Mbps

Access networks: data center networks

- high-bandwidth links (10s to 100s Gbps) connect hundreds to thousands of servers together, and to Internet



Courtesy: Massachusetts Green High Performance Computing Center (mgppcc.org)



Data Centers

- Enterprise/University Network.



Data Center Tiers

- Tier 1 = Non-redundant capacity components (single uplink and servers).
- Tier 2 = Tier 1 + Redundant capacity components.
- Tier 3 = Tier 1 + Tier 2 + Dual-powered equipment and multiple uplinks.
- Tier 4 = Tier 1 + Tier 2 + Tier 3 + all components are fully fault-tolerant including uplinks, storage, chillers, HVAC systems (heating, ventilation, and air conditioning), servers etc. Everything is dual-powered.
- Data Center Availability According To Tiers
- The levels also describes the availability of data from the hardware at a location as follows:
 - Tier 1: Guaranteeing 99.671% availability.
 - Tier 2: Guaranteeing 99.741% availability.
 - Tier 3: Guaranteeing 99.982% availability.
 - Tier 4: Guaranteeing 99.995% availability

Links: physical media

- **bit**: propagates between transmitter/receiver pairs
- **physical link**: what lies between transmitter & receiver
- **guided media**:
 - signals propagate in solid media: copper, fiber, coax
- **unguided media**:
 - signals propagate freely, e.g., radio

Twisted pair (TP)

- two insulated copper wires
 - Category 5: 100 Mbps, 1 Gbps Ethernet
 - Category 6: 10Gbps Ethernet



Links: physical media

Coaxial cable:

- two concentric copper conductors
- bidirectional
- broadband:
 - multiple frequency channels on cable
 - 100's Mbps per channel



Fiber optic cable:

- glass fiber carrying light pulses, each pulse a bit
- high-speed operation:
 - high-speed point-to-point transmission (10's-100's Gbps)
- low error rate:
 - repeaters spaced far apart
 - immune to electromagnetic noise



Links: physical media

Wireless radio

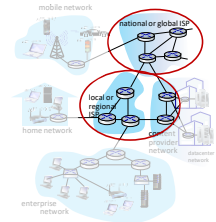
- signal carried in various "bands" in electromagnetic spectrum
- no physical "wire"
- broadcast, "half-duplex" (sender to receiver)
- propagation environment effects:
 - reflection
 - obstruction by objects
 - interference/noise

Radio link types:

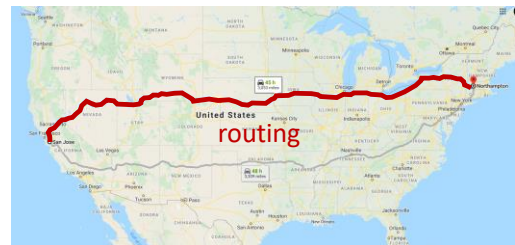
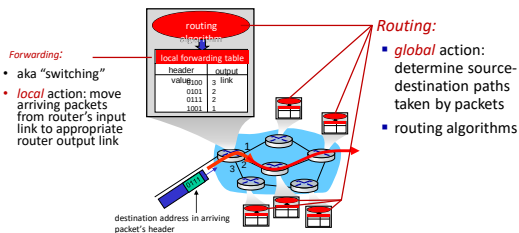
- **Wireless LAN (WiFi)**
 - 10-100's Mbps; 100's of meters
- **wide-area** (e.g., 4G cellular)
 - 10's Mbps over ~10 Km
- **Bluetooth**: cable replacement
 - short distances, limited rates
- **terrestrial microwave**
 - point-to-point; 45 Mbps channels
- **satellite**
 - up to 45 Mbps per channel
 - 270 msec end-end delay

The network core

- mesh of interconnected routers
- **packet-switching**: hosts break application-layer messages into **packets**
 - network **forwards** packets from one router to the next, across links on path from **source** to **destination**



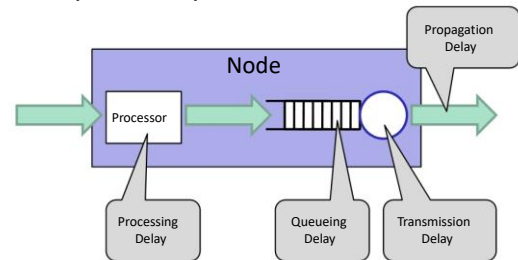
Two key network-core functions



Sources of Delay

- **Processing Delay**
 - Time between when a packet is received and scheduled on an outgoing queue
- **Queuing Delay**
 - Time packet spends in the outgoing queue
- **Transmission Delay**
 - Time between when the first and last bits are transmitted
- **Propagation Delay**
 - Time it takes for the last bit to leave the transmitter and arrive at the receiver

Graphical Representation



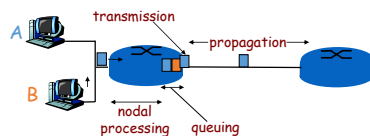
Delay in packet-switched networks

nodal processing:

- check bit errors
- determine output link

queuing

- time waiting at output link for transmission
- depends on congestion level of router



Delay in packet-switched networks

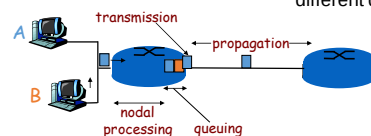
Transmission delay:

- R = link bandwidth (bps)
- L = packet length (bits)
- time to send bits into link = L/R

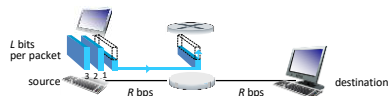
Propagation delay:

- d = length of physical link
- s = propagation speed in medium ($\sim 2 \times 10^8$ m/sec)
- propagation delay = d/s

Note: s and R are very different quantities!



Packet-switching: store-and-forward

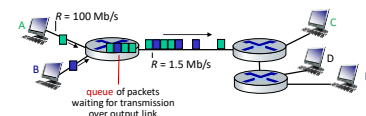


- **packet transmission delay:** takes L/R seconds to transmit (push out) L -bit packet into link at R bps
- **store and forward:** entire packet must arrive at router before it can be transmitted on next link

One-hop numerical example:

- $L = 10$ Kbits
- $R = 100$ Mbps
- one-hop transmission delay = 0.1 msec

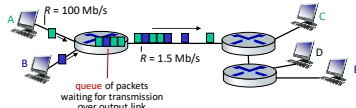
Packet-switching: queueing



Queueing occurs when work arrives faster than it can be serviced:



Packet-switching: queueing

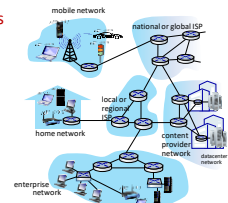


Packet queuing and loss: if arrival rate (in bps) to link exceeds transmission rate (bps) of link for some period of time:

- packets will queue, waiting to be transmitted on output link
- packets can be dropped (lost) if memory (buffer) in router fills up

Internet structure: a “network of networks”

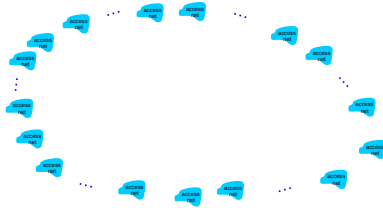
- hosts connect to Internet via **access** Internet Service Providers (ISPs)
- access ISPs in turn must be interconnected
 - so that *any* two hosts (*anywhere!*) can send packets to each other
- resulting network of networks is very complex
 - evolution driven by **economics**, **national policies**



Let's take a stepwise approach to describe current Internet structure

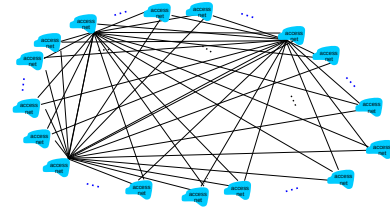
Internet structure: a “network of networks”

Question: given *millions* of access ISPs, how to connect them together?



Internet structure: a “network of networks”

Question: given *millions* of access ISPs, how to connect them together?



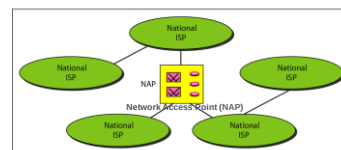
Internet Service Providers (ISPs)

- **Internet today**
 - Made up of many wide and local area networks joined by connecting devices and switching stations. Today most end users use the **services of internet service providers (ISPs)**.
- Hierarchical organization of the Internet includes:
 - International Internet Service Providers (Interconnection of National ISPs)
 - National Internet Service Providers
 - Regional Internet Service Providers
 - Local Internet Service Providers

Internet Service Providers (ISPs)



a. Structure of a national ISP



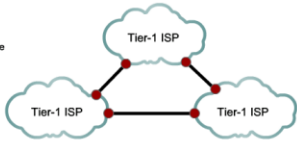
b. Interconnection of national ISPs

Internet

Internet Structure - A Network of Networks

At the center of the Internet, "tier-1" ISPs provide national and international connections. These ISPs treat each other as equals.

Examples are: Verizon, Sprint, AT&T, NTT, cable systems and wide area wireless networks.

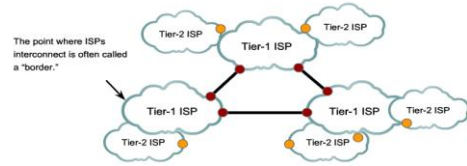


This is the "backbone" of the Internet.

Internet

Internet Structure - A Network of Networks

"Tier-2" ISPs are smaller and often provide regional service. Tier-2 ISPs usually pay Tier-1 ISPs for connectivity to rest of the Internet.

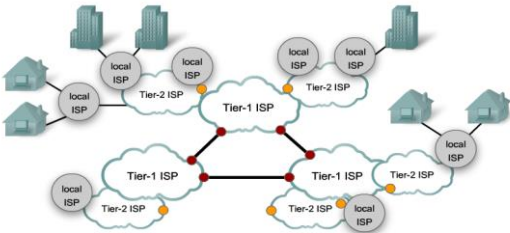


The point where ISPs interconnect is often called a "border."

Internet

Internet Structure - A Network of Networks

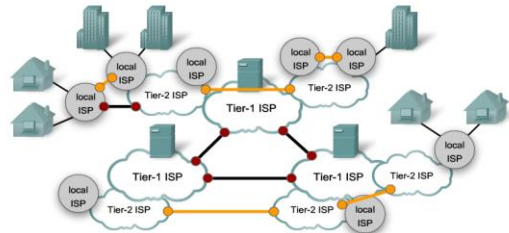
"Tier-3" ISPs are the local providers of service directly to end users. Tier-3 ISPs are usually connected to Tier 2 ISPs and pay Tier 2 providers for Internet access.



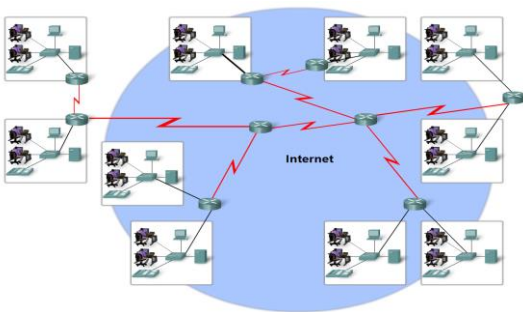
Internet

Internet Structure - A Network of Networks

Peer connections between networks at the same level provide direct connections, bypassing longer routes and preventing congestion on the backbone.

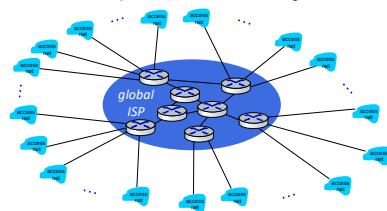


Internet



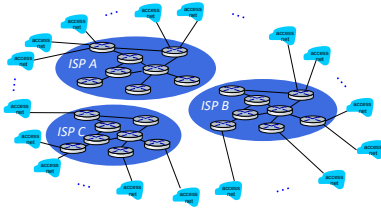
Internet structure: a "network of networks"

*Option: connect each access ISP to one global transit ISP?
Customer and provider ISPs have economic agreement.*



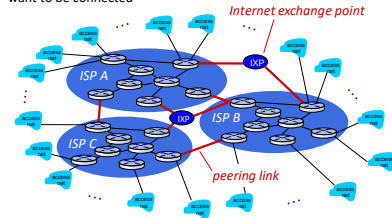
Internet structure: a “network of networks”

But if one global ISP is viable business, there will be competitors



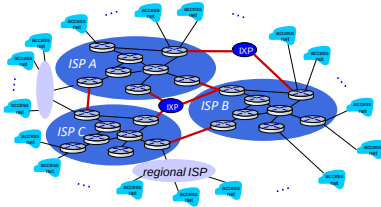
Internet structure: a “network of networks”

But if one global ISP is viable business, there will be competitors who will want to be connected



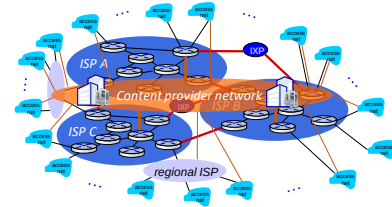
Internet structure: a “network of networks”

... and regional networks may arise to connect access nets to ISPs

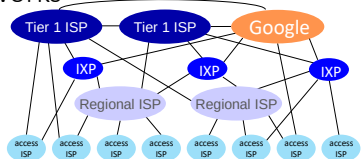


Internet structure: a “network of networks”

... and content provider networks (e.g., Google, Microsoft, Akamai) may run their own network, to bring services, content close to end users



Internet structure: a “network of networks”



At “center”: small # of well-connected large networks

- “tier-1” commercial ISPs (e.g., Level 3, Sprint, AT&T, NTT), national & international coverage
- content provider networks (e.g., Google, Facebook): private network that connects its data centers to Internet, often bypassing tier-1, regional ISPs

Lines of defense:

- **authentication:** proving you are who you say you are
 - cellular networks provides hardware identity via SIM card; no such hardware assist in traditional Internet
- **confidentiality:** via encryption
- **integrity checks:** digital signatures prevent/detect tampering
- **access restrictions:** password-protected VPNs
- **firewalls:** specialized “middleboxes” in access and core networks:
 - off-by-default: filter incoming packets to restrict senders, receivers, applications
 - detecting/reacting to DOS attacks

... lots more on security (throughout, Chapter 8)

Internet history

1961-1972: Early packet-switching principles

- 1961: Kleinrock - queueing theory shows effectiveness of packet-switching
- 1964: Baran - packet-switching in military nets
- 1967: ARPAnet conceived by Advanced Research Projects Agency
- 1969: first ARPAnet node operational
- 1972:
 - ARPAnet public demo
 - NCP (Network Control Protocol) first host-host protocol
 - first e-mail program
 - ARPAnet has 15 nodes



Internet history

1972-1980: Internetworking, new and proprietary networks

- 1970: ALOHAnet satellite network in Hawaii
- 1974: Cerf and Kahn - architecture for interconnecting networks
- 1976: Ethernet at Xerox PARC
- late 70's: proprietary architectures: DECnet, SNA, XNA
- 1979: ARPAnet has 200 nodes

Cerf and Kahn's internetworking principles:

- minimalism, autonomy - no internal changes required to interconnect networks
- best-effort service model
- stateless routing
- decentralized control

define today's Internet architecture

Internet history

1990, 2000s: commercialization, the Web, new applications

- early 1990s: ARPAnet decommissioned
- 1991: NSF lifts restrictions on commercial use of NSFnet (decommissioned, 1995)
- early 1990s: Web
 - hypertext [Bush 1945, Nelson 1960's]
 - HTML, HTTP: Berners-Lee
 - 1994: Mosaic, later Netscape
 - late 1990s: commercialization of the Web
- late 1990s – 2000s:
 - more killer apps: instant messaging, P2P file sharing
 - network security to forefront
 - est. 50 million host, 100 million+ users
 - backbone links running at Gbps

Internet history

2005-present: scale, SDN, mobility, cloud

- aggressive deployment of broadband home access (10-100's Mbps)
- 2008: software-defined networking (SDN)
- increasing ubiquity of high-speed wireless access: 4G/5G, WiFi
- service providers (Google, FB, Microsoft) create their own networks
 - bypass commercial Internet to connect "close" to end user, providing "instantaneous" access to social media, search, video content, ...
- enterprises run their services in "cloud" (e.g., Amazon Web Services, Microsoft Azure)
- rise of smartphones: more mobile than fixed devices on Internet (2017)
- ~18B devices attached to Internet (2017)

Summary

We've covered a "ton" of material!

- Internet overview
- what's a protocol?
- network edge, access network, core
 - packet-switching versus circuit-switching
 - Internet structure
- performance: loss, delay, throughput
- layering, service models
- security
- history

You now have:

- context, overview, vocabulary, "feel" of networking
- more depth, detail, and fun to follow!